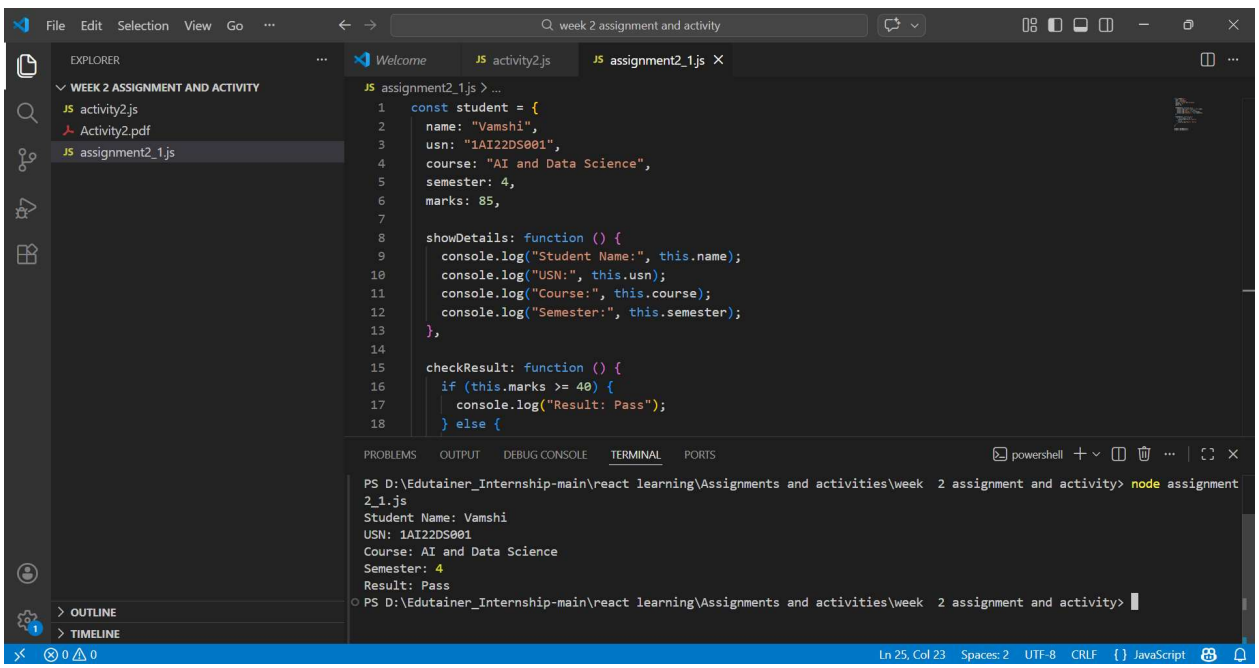
Title: Object-Oriented JavaScript Model

Description: Design a real-world JavaScript object (student, product, employee) with properties and methods using object literals and this.

Reference/Source Code/Links:
https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide/Working_with_Objects
https://www.w3schools.com/js/js_objects.asp

Deliverable:
JavaScript object file
Method execution outputs
Explanation of object design

Student object:-



Explanation of object design:-

In this program, a student object is created using object literal syntax {}.The object contains properties and methods.

Properties

Properties store data about the student.

- name – student name
- usn – student unique ID

- course – student course name
- semester – current semester of the student
- marks – marks obtained by the student

Methods

Methods are functions inside objects that perform actions.

- showDetails() – prints student information
- checkResult() – checks and prints whether the student passed or failed

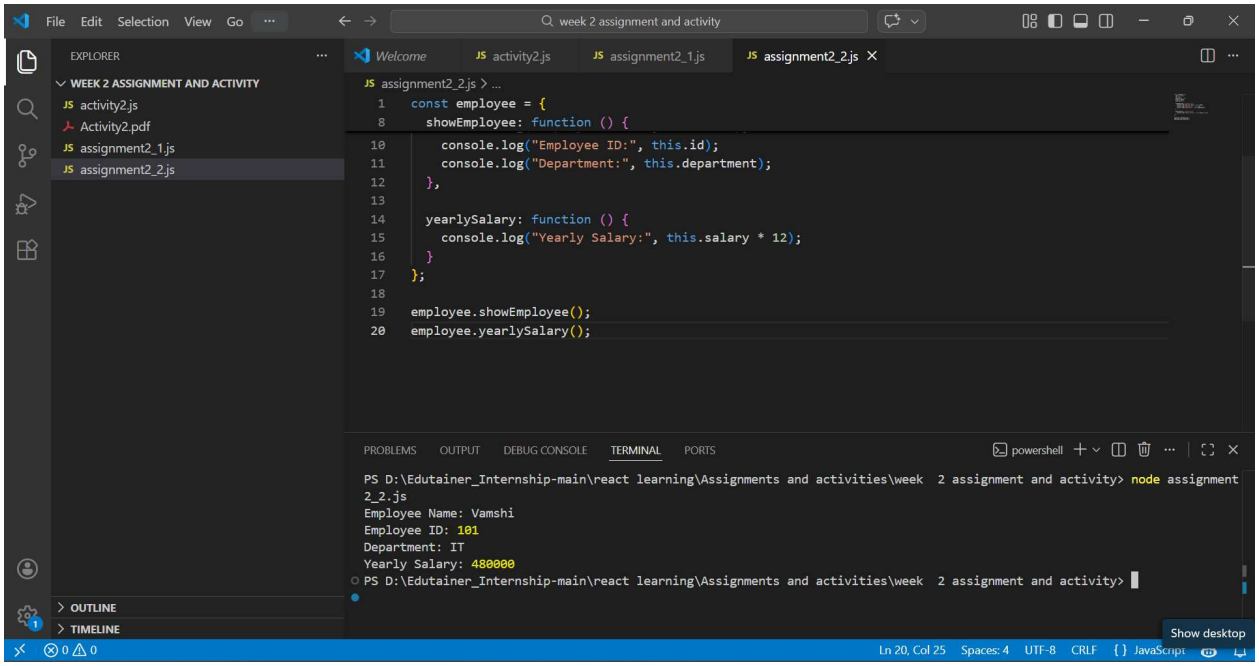
Use of this

The this keyword refers to the current object.

Example:

this.name refers to the name property of the student object.

Employee object:-



In this program, an employee object is created using object literal syntax {}.

The object contains properties and methods.

Properties

Properties store data about the employee.

- name – employee name
- id – employee identification number
- department – employee department
- salary – monthly salary
- experience – years of experience

Methods

Methods are functions inside objects that perform actions.

- showEmployee() – prints employee information
- yearlySalary() – calculates and prints yearly salary

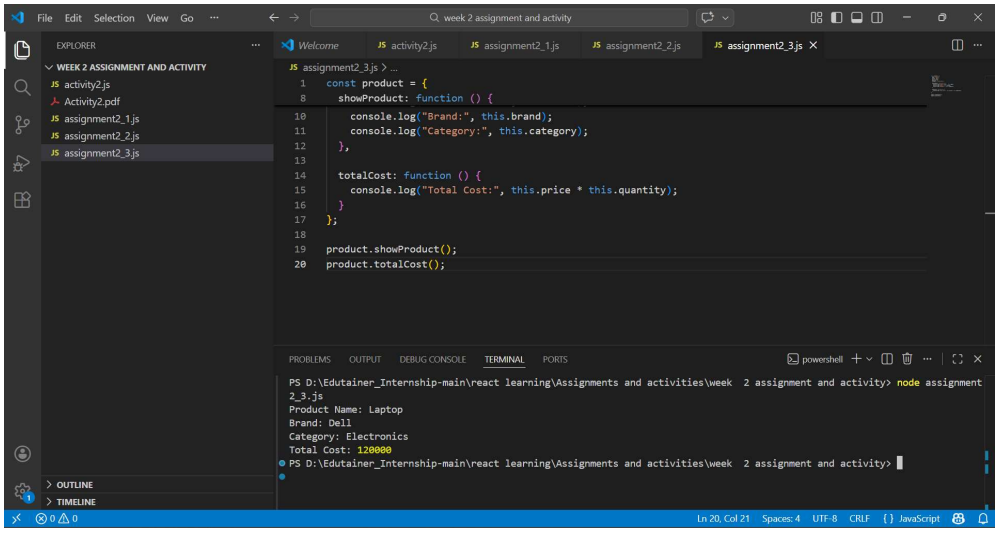
Use of this

The this keyword refers to the current object.

Example:

this.salary refers to the salary property of the employee object.

Product object



In this program, a product object is created using object literal syntax {}.

The object contains properties and methods.

Properties

Properties store data about the product.

- name – product name
- brand – brand of the product
- price – price of the product
- quantity – number of products available
- category – product category

Methods

Methods are functions inside objects that perform actions.

- showProduct() – prints product details
- totalCost() – calculates and prints the total cost of the product

Use of this

The this keyword refers to the current object.

Example:

this.price refers to the price property of the product object.